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2º ASIR

Tarea. Destilación

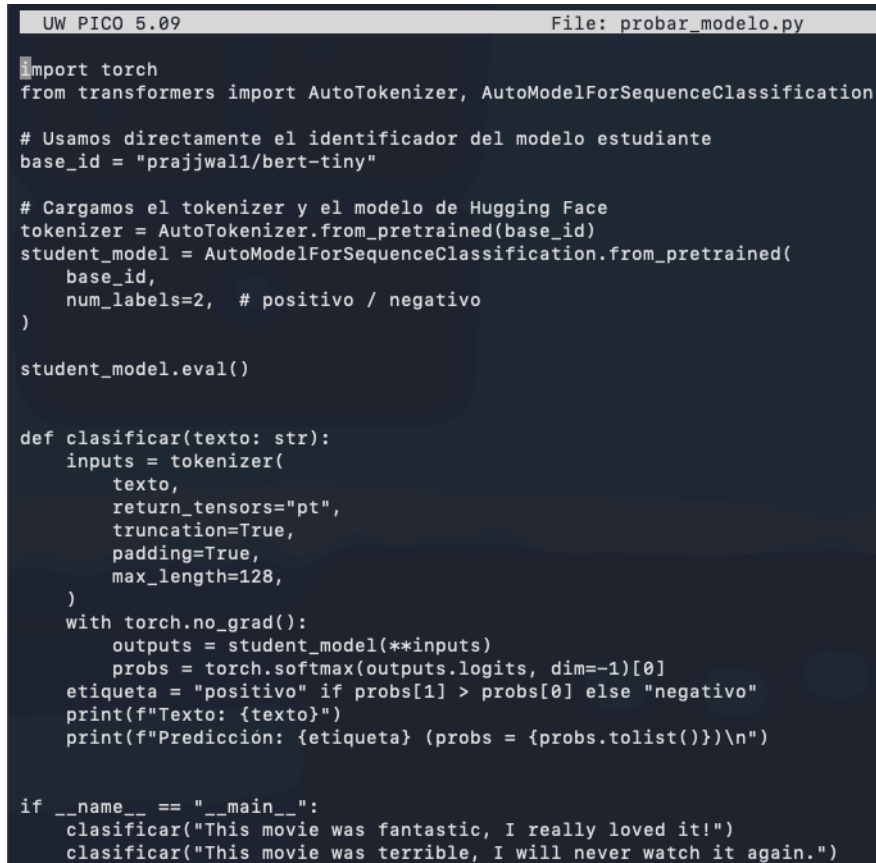


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1. Script de entrenamiento.

- **Modelo Profesor:** “distilbert-base-uncased-finetuned-sst-2” (un modelo experto en análisis de sentimientos).
- **Modelo Estudiante:** “bert-tiny” (un modelo más pequeño y rápido, inicialmente sin entrenar).
- **Dataset:** Se utilizó el dataset SST-2, que tiene frases calificadas como positivas o negativas.



```
UW PICO 5.09 File: probar_modelo.py

import torch
from transformers import AutoTokenizer, AutoModelForSequenceClassification

# Usamos directamente el identificador del modelo estudiante
base_id = "prajjwall/bert-tiny"

# Cargamos el tokenizer y el modelo de Hugging Face
tokenizer = AutoTokenizer.from_pretrained(base_id)
student_model = AutoModelForSequenceClassification.from_pretrained(
    base_id,
    num_labels=2, # positivo / negativo
)

student_model.eval()

def clasificar(texto: str):
    inputs = tokenizer(
        texto,
        return_tensors="pt",
        truncation=True,
        padding=True,
        max_length=128,
    )
    with torch.no_grad():
        outputs = student_model(**inputs)
        probs = torch.softmax(outputs.logits, dim=-1)[0]
        etiqueta = "positivo" if probs[1] > probs[0] else "negativo"
        print(f"Texto: {texto}")
        print(f"Predicción: {etiqueta} (probs = {probs.tolist()})\n")

if __name__ == "__main__":
    clasificar("This movie was fantastic, I really loved it!")
    clasificar("This movie was terrible, I will never watch it again.")
```

2. Proceso de Destilación.

Ejecución del script para que el modelo estudiante aprendiese de las predicciones del modelo profesor.

- Se activó el entorno virtual y se lanzó python [destilar.py](#).

- Al cargar "bert-tiny", aparecen mensajes de "MISSING" o "UNEXPECTED", pero es normal porque el modelo se está adaptando a la tarea nueva de clasificación.
- El entrenamiento se configuró para 1000 iteraciones. Se puede ver el progreso con una pérdida (loss) de 1.934.

```
UW PICO 5.09 File: destilar.py

import torch
import torch.nn.functional as F

from datasets import load_dataset
from transformers import (
    AutoTokenizer,
    AutoModelForSequenceClassification,
    TrainingArguments,
    Trainer,
)
from sklearn.metrics import accuracy_score, f1_score

# -----
# 1. Cargar modelos y tokenizer
# -----
teacher_id = "distilbert/distilbert-base-uncased-finetuned-sst-2-english"
student_id = "prajjwall/bert-tiny"

teacher_model = AutoModelForSequenceClassification.from_pretrained(teacher_id)
student_model = AutoModelForSequenceClassification.from_pretrained(
    student_id,
    num_labels=2, # positivo / negativo
)

tokenizer = AutoTokenizer.from_pretrained(student_id)

# -----
# 2. Cargar y tokenizar dataset
# -----
dataset = load_dataset("glue", "sst2")

def tokenize_data(examples):
    return tokenizer(
        examples["sentence"],
        truncation=True,
        padding="max_length",
        max_length=128,
    )

tokenized_dataset = dataset.map(tokenize_data, batched=True)

# Hugging Face Trainer espera "labels"
tokenized_dataset = tokenized_dataset.rename_column("label", "labels")
[Wrote 149 lines]
```

```
(.venv) aulaateca26@Mac-mini-de-aulaateca26 destilacion-ia % cd /Users/aulaateca26/2asir/destilacion-ia
a
source .venv/bin/activate
python destilar.py

Loading weights: 100%|█ 104/104 [00:00<00:00, 5839.07it/s, Materializing param=pre_classifier.weight]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%|█ 39/39 [00:00<00:00, 9090.69it/s, Materializing param=bert.pooler.dense.weight]
BertForSequenceClassification LOAD REPORT from: prajjwall/bert-tiny
Key | Status
-----|-----
cls.seq_relationship.weight | UNEXPECTED
cls.predictions.transform.dense.bias | UNEXPECTED
cls.predictions.transform.LayerNorm.weight | UNEXPECTED
cls.predictions.transform.dense.weight | UNEXPECTED
cls.predictions.decoder.bias | UNEXPECTED
cls.seq_relationship.bias | UNEXPECTED
cls.predictions.decoder.weight | UNEXPECTED
cls.predictions.transform.LayerNorm.bias | UNEXPECTED
cls.predictions.bias | UNEXPECTED
bert.embeddings.position_ids | UNEXPECTED
classifier.bias | MISSING
classifier.weight | MISSING

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
Iniciando la destilación (entrenamiento recortado)...
0%| 0/1000 [00:00<?, ?it/s]
/Users/aulaateca26/2asir/destilacion-ia/.venv/lib/python3.14/site-packages/torch/utils/data/dataloader.py:775: UserWarning: 'pin_memory' argument is set as true but not supported on MPS now, device pinned memory won't be used.
  super().__init__(loader)
{'loss': '1.934', 'grad_norm': '1.703', 'learning_rate': '4.755e-05', 'epoch': '0.1597'}
5%| 54/1000 [00:09<02:42, 5.84it/s]
```

3. Prueba.

Se crea un script que carga el modelo estudiante y define una función “clasificar” para que devuelva la probabilidad de que una frase sea positiva o negativa. Los resultados son los siguientes:

- "This movie was fantastic, I really loved it!" -> Predicción: positivo (60% de confianza).
- "This movie was terrible..." -> Predicción: positivo (60% de confianza).

Estos resultados demuestran que el modelo todavía necesita más entrenamiento porque clasifica las dos frases como positivas.

```
(.venv) aulaateca26@Mac-mini-de-aulaateca26 destilacion-ia % cd /Users/aulaateca26/2asir/destilacion-ia
source .venv/bin/activate
python probar_modelo.py

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
Loading weights: 100%|█ 39/39 [00:00<00:00, 11080.26it/s, Materializing param=bert.pooler.dense.weight]
BertForSequenceClassification LOAD REPORT from: prajjwall1/bert-tiny
-----
Key | Status |
-----+-----+
cls.predictions.transform.dense.bias | UNEXPECTED |
cls.predictions.decoder.bias | UNEXPECTED |
cls.seq_relationship.weight | UNEXPECTED |
cls.predictions.bias | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
cls.predictions.decoder.weight | UNEXPECTED |
bert.embeddings.position_ids | UNEXPECTED |
cls.predictions.transform.dense.weight | UNEXPECTED |
cls.seq_relationship.bias | UNEXPECTED |
classifier.weight | MISSING |
classifier.bias | MISSING |
-----

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
Texto: This movie was fantastic, I really loved it!
Predicción: positivo (probs = [0.3990715742111206, 0.6009284257888794])

Texto: This movie was terrible, I will never watch it again.
Predicción: positivo (probs = [0.3918722867965698, 0.6081277132034302])

(.venv) aulaateca26@Mac-mini-de-aulaateca26 destilacion-ia %
```

4. Comparativa de rendimiento.

- **Rendimiento del Profesor:** Precisión de 0.9106 (91%) y un resultado F1 de 0.9105.
- **Rendimiento del Estudiante:** Precisión de 0.4874 (48%).

Hay una diferencia de 0.4232 en la precisión lo que significa que, aunque el estudiante es mucho más ligero, claramente necesita un proceso de destilación bastante más largo y completo para alcanzar al profesor.

Métrica	Modelo Profesor (DistilBERT)	Modelo Estudiante (Bert-tiny)
Precisión	91.06%	48.74%
Resultado F1	0.9105	0.3390

```
destilacion-ia -- nano comparar_profesor_estudiante.py -- 102x50
UW PICO 5.09 File: comparar_profesor_estudiante.py

import torch
from datasets import load_dataset
from transformers import (
    AutoTokenizer,
    AutoModelForSequenceClassification,
)
from sklearn.metrics import accuracy_score, f1_score

# IDs de los modelos
teacher_id = "distilbert/distilbert-base-uncased-finetuned-sst-2-english"
student_id = "prajjwal1/bert-tiny"

# Ruta al modelo destilado (salida de destilar.py)
student_path = "distilled_model" # si no tienes esta carpeta, usa student_id directamente

# 1. Cargar modelos
print("Cargando modelo PROFESOR...")
teacher_model = AutoModelForSequenceClassification.from_pretrained(teacher_id)
teacher_model.eval()

print("Cargando modelo ESTUDIANTE...")
try:
    # Intentamos cargar el modelo destilado si existe
    student_model = AutoModelForSequenceClassification.from_pretrained(
        student_path,
        num_labels=2,
    )
    print("Usando modelo estudiante destilado desde 'distilled_model'.")
except Exception as e:
    print("No se pudo cargar 'distilled_model'. Usando prajjwal1/bert-tiny sin destilar.")
    print("Detalle del error:", e)
    student_model = AutoModelForSequenceClassification.from_pretrained(
        student_id,
        num_labels=2,
    )
student_model.eval()

# Tokenizador (mismo tamaño de vocabulario que el estudiante)
tokenizer = AutoTokenizer.from_pretrained(student_id)

# 2. Cargar dataset de validación
print("Cargando dataset SST-2 (validación)...")

Get Help WriteOut Read File Prev Pg Cut Text Cur Pos
Exit Justify Where is Next Pg UnCut Text To Spell
```

```
destilacion-ia -- zsh -- 102x50

formerConfig, LukeConfig, MarkuPLMConfig, MBartConfig, MegatronBertConfig, MiniMaxConfig, MinistralCon
fig, Ministral3Config, MistralConfig, MixtralConfig, MobileBertConfig, ModernBertConfig, ModernBertDec
oderConfig, MPNetConfig, MptConfig, MraConfig, MT5Config, MvpConfig, NemotronConfig, NystromformerCon
fig, OpenAIIGPTConfig, OPTConfig, PerceiverConfig, PersimmonConfig, PhiConfig, Phi3Config, PhimoeConfig,
PLBartConfig, Qwen2Config, Qwen2MoeConfig, Qwen3Config, Qwen3MoeConfig, Qwen3NextConfig, ReformerCon
fig, RemBertConfig, RobertaConfig, RobertaPreLayerNormConfig, RoCBertConfig, RoFormerConfig, SeedOssCon
fig, SmoLLM3Config, SqueezeBertConfig, StableLmConfig, Starcoder2Config, T5Config, T5GemmaConfig, T5Ge
mma2Config, TapasConfig, UMT5Config, XLMConfig, XLMRobertaConfig, XLMRobertaXLConfig, XLNetConfig, Xmo
dConfig, YosoConfig, ZambaConfig, Zamba2Config.
Loading weights: 100%| 39/39 [00:00<00:00, 8383.45it/s, Materializing param=bert.pooler.dense.weight
BertForSequenceClassification LOAD REPORT from: prajjwal1/bert-tiny
Key | Status |
-----|-----|
cls.seq_relationship.weight | UNEXPECTED |
cls.predictions.transform.LayerNorm.bias | UNEXPECTED |
cls.predictions.transform.dense.weight | UNEXPECTED |
cls.seq_relationship.bias | UNEXPECTED |
cls.predictions.decoder.weight | UNEXPECTED |
cls.predictions.transform.LayerNorm.weight | UNEXPECTED |
cls.predictions.decoder.bias | UNEXPECTED |
bert.embeddings.position_ids | UNEXPECTED |
cls.predictions.bias | UNEXPECTED |
cls.predictions.transform.dense.bias | UNEXPECTED |
classifier.weight | MISSING |
classifier.bias | MISSING |

Notes:
- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect id
entical arch.
- MISSING :those params were newly initialized because missing from the checkpoint. Consider tra
ining on your downstream task.
Cargando dataset SST-2 (validación)...
Map: 100%| 872/872 [00:00<00:00, 27999.06 examples/s]

Resultados para PROFESOR (DistilBERT):
Accuracy: 0.9186
F1-score: 0.9185

Resultados para ESTUDIANTE (BERT-tiny destilado):
Accuracy: 0.4874
F1-score: 0.3390

--- COMPARACIÓN PROFESOR vs ESTUDIANTE ---
Profesor - Accuracy: 0.9186 | F1: 0.9185
Estudiante- Accuracy: 0.4874 | F1: 0.3390
Diferencia de accuracy: 0.4232
Diferencia de F1: 0.5715
(.venv) aulaateca26@Mac-mini-de-aulaateca26 destilacion-ia %
```